

Lesson 05 Practice: Answers and Rationales

JKSSB Computer Awareness

October 10, 2026

Provenance: All 25 items are original exam-pattern questions (*[Original]* plus the *[JKSSB FAA-pattern]*, *[JKPSI-pattern]*, *[SSC CGL-pattern]*, and *[JKSSB-pattern]* extension items, including MSQs); none reproduces an official paper item.

- Q1. A.** ALU stands for Arithmetic Logic Unit. B, C, and D are not expansions of the abbreviation.
- Q2. C.** The Arithmetic Logic Unit performs arithmetic and logical operations. The Control Unit directs operations; the register file stores values; the system bus carries signals.
- Q3. B.** The Control Unit directs the sequence of operations and issues control signals. Performing arithmetic is the ALU's job (A); the CU does not store results (C) or provide secondary storage (D).
- Q4. D.** Adding two numbers is an arithmetic operation performed by the ALU, not the Control Unit. Decoding instructions, issuing control signals, and directing data flow are all Control Unit functions.
- Q5. B.** The Program Counter holds the address of the next instruction to be fetched. The IR holds the current instruction; the MDR holds data; the accumulator holds intermediate results.
- Q6. C.** The Instruction Register holds the instruction currently being decoded and executed. The PC holds the next address; the MDR holds data; the MAR holds an address.
- Q7. B.** The Memory Address Register holds the address of the memory location to be accessed. The MDR holds data; the IR holds an instruction; the accumulator holds a result.
- Q8. C.** The Memory Data Register holds the data being transferred to or from memory. The MAR holds the address; the PC holds the next instruction address; the IR holds the instruction.
- Q9. B.** The accumulator holds intermediate arithmetic and logic results. It does not hold the next-instruction address (A), the fetched instruction (C), or a permanent copy of the OS (D).
- Q10. A.** Registers are the fastest storage, followed by L1 cache, RAM, and secondary storage such as an SSD. B, C, and D are progressively slower.
- Q11. D.** Registers are located inside the CPU. They are not part of cache (A), main memory (B), or secondary storage (C).
- Q12. B.** One gigahertz is one billion (10^9) cycles per second, so 2.4 GHz is about 2.4 billion cycles per second. A, C, and D misstate the giga- scale.
- Q13. C.** Different instructions may take different numbers of clock cycles, so A and B are false. A clock cycle is a unit of time, not bytes (D).

- Q14. B.** Statements I and II are correct: the ALU works on binary data with logic gates, and the Control Unit directs operations and data flow. III is false because the Control Unit does not perform arithmetic. A omits II; C includes false III and omits I; D includes false III.
- Q15. A.** PC is correctly matched with Program Counter. IR is Instruction Register, MAR is Memory Address Register, and MDR is Memory Data Register, so B, C, and D are wrong expansions.
- Q16. A.** The Program Counter holds the address of the next instruction to be fetched. It does not hold the current instruction (B), the previously executed one (C), or data (D).
- Q17. B.** Registers with no fixed dedicated role, used for general operands and intermediate results, are general-purpose registers. Status registers hold flags; MAR holds addresses; IR holds instructions.
- Q18. C.** The status or flags register holds condition flags such as zero, carry, and sign. A describes the MAR; B describes the IR; D describes the MDR.
- Q19. D.** A quad-core processor contains four independent processing cores on a single chip. It does not mean four bytes of cache (A), four monitors (B), or four operating systems (C).
- Q20. B.** The Control Unit fetches each instruction, decodes it, and issues control signals to the other units. The ALU computes; the hard disk is secondary storage; the monitor is an output device.
- Q21. A.** The Program Counter tracks the next instruction. B, C, and D hold other values, not the next-instruction address.
- Q22. A.** MAR is the Memory Address Register. B, C, and D are not the recognised expansion.
- Q23. A.** Comparisons are logical operations done by the ALU. B, C, and D are peripherals or storage.
- Q24. A, B, C.** The PC, IR, and MAR are registers. A USB port (D) is an interface, not a register.
- Q25. A, B, C.** The PC, IR, and ALU roles are stated correctly. D is false: the CU directs work; it does not store files.
- Q26. C.** The Instruction Register holds the fetched instruction during decode/execute. MAR holds an address, PC holds the next instruction address, and the accumulator stores intermediate results.
- Q27. B.** The Program Counter holds the next instruction address. The IR holds the instruction itself, the accumulator holds results, and MDR holds data transferred to/from memory.
- Q28. A.** The Memory Address Register holds a memory address. IR holds an instruction, the accumulator stores results, and a status register stores condition flags.
- Q29. B.** Clock rate is cycles per second, measured in hertz. The other units measure data rate, image density, and rotational speed.
- Q30. A, B, C, D.** All four register-role pairings are correct.